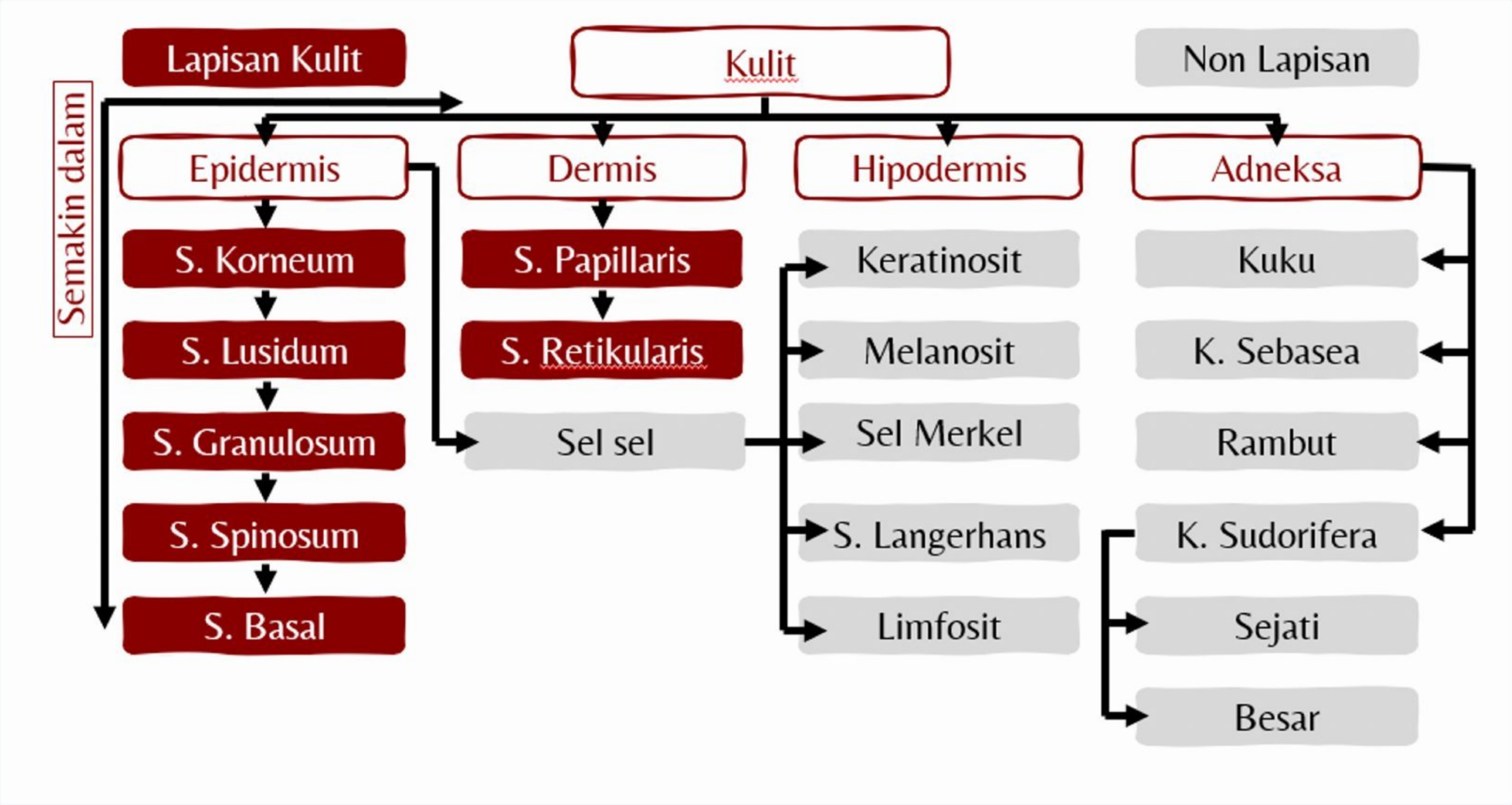
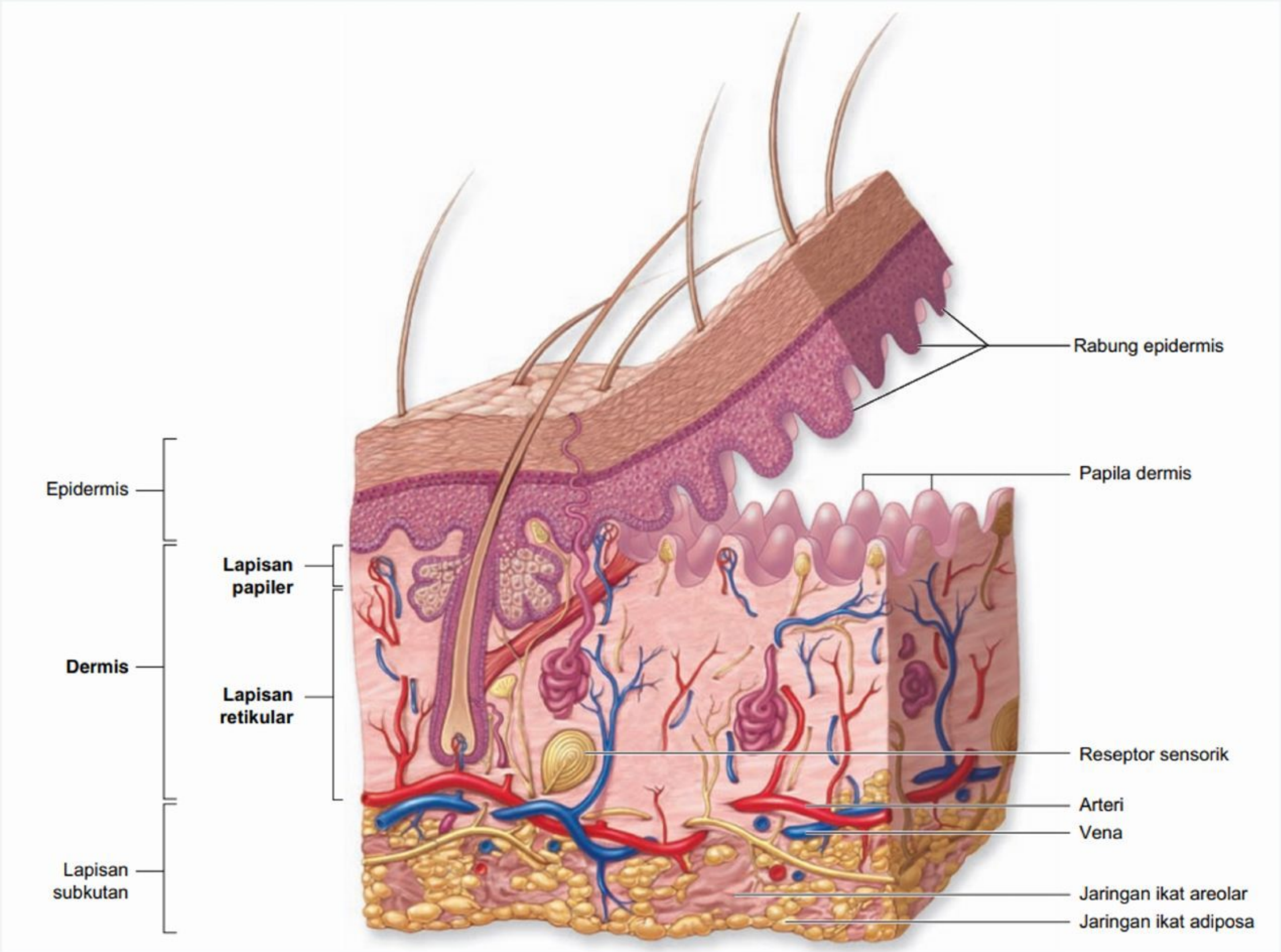


SKIN STRUCTURE

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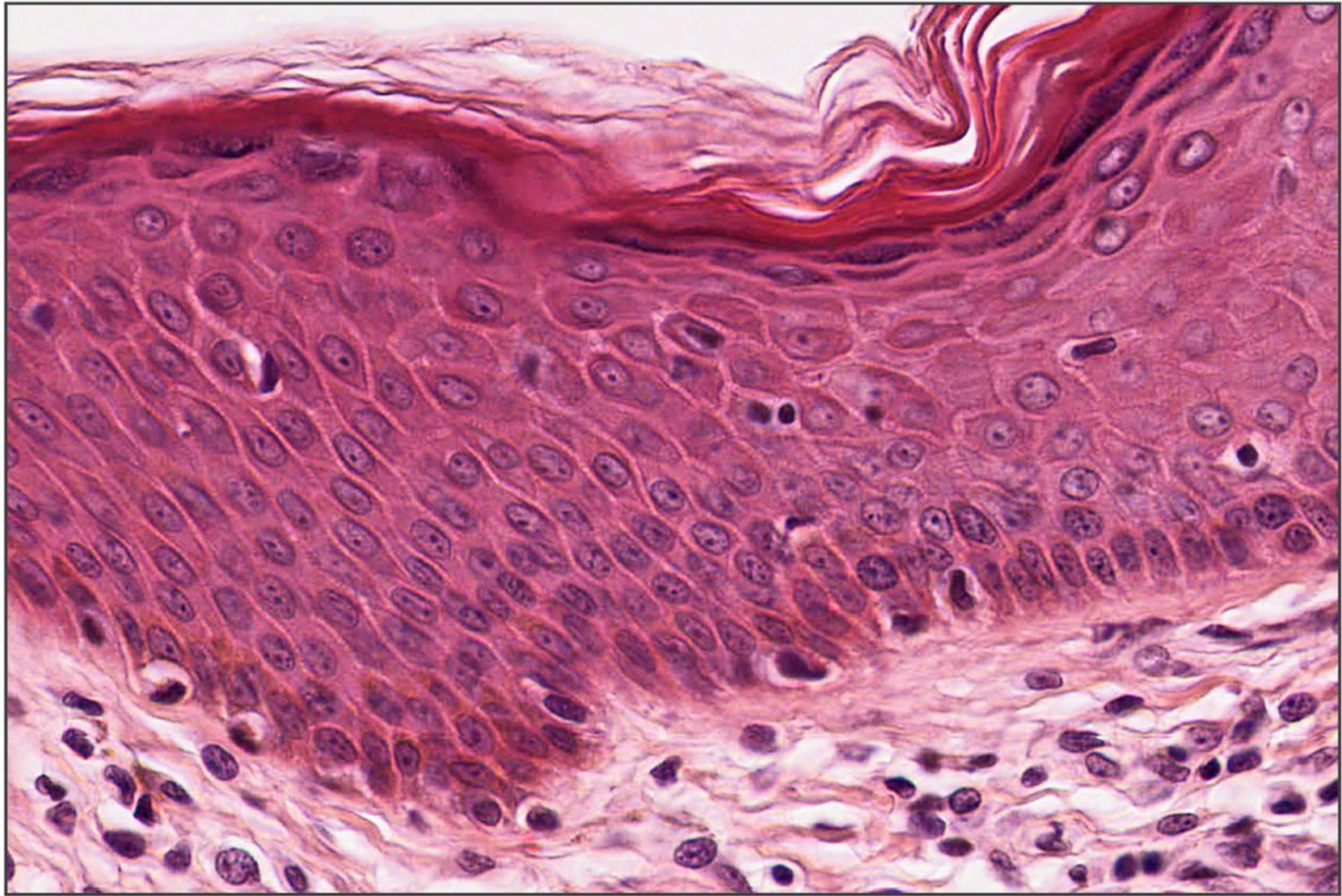




EPIDERMIS

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EPIDERMIS





EPIDERMIS

The outermost layer of the skin composed of
keratinized stratified squamous epithelium.

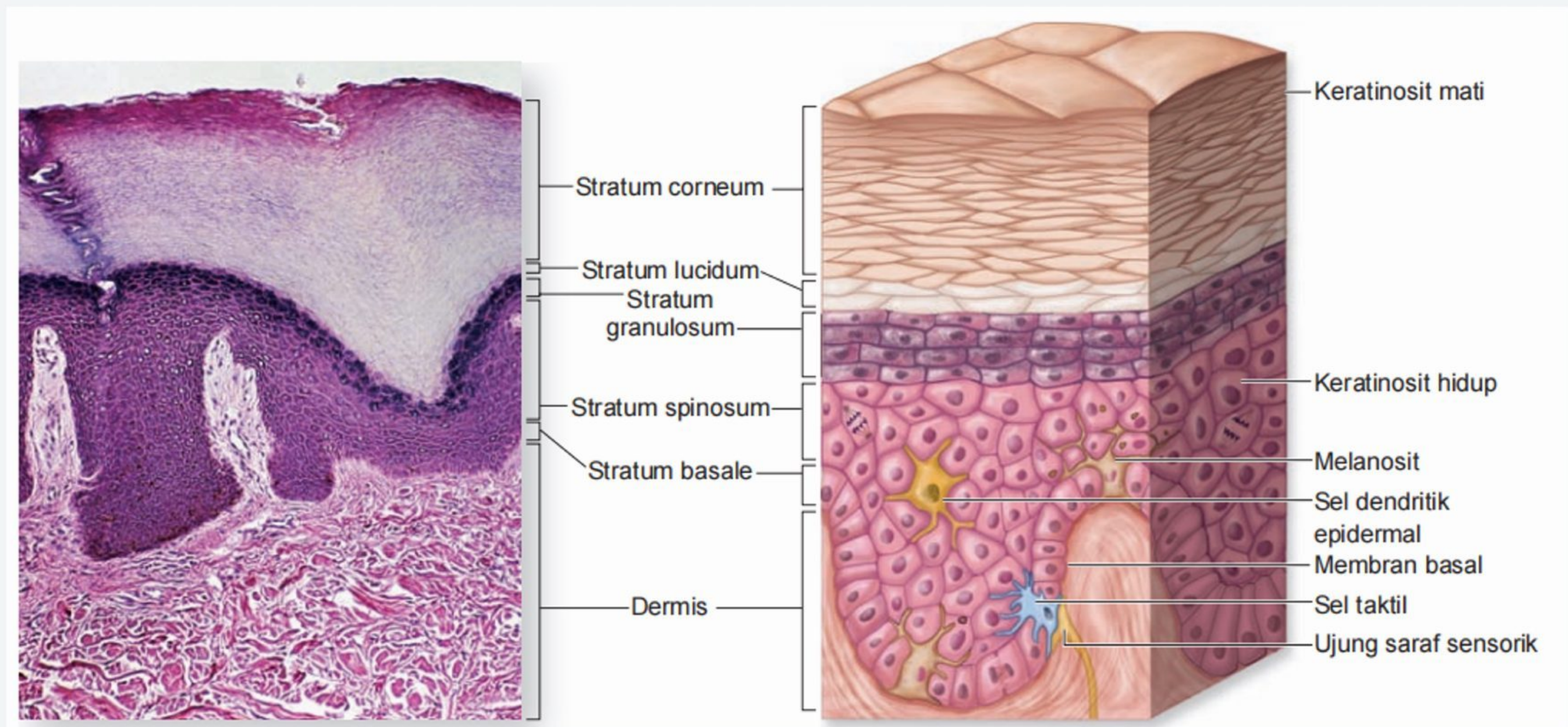
THIN SKIN

- Stratum Korneum
- Stratum Granulosum
- Stratum Spinosum
- Stratum Basalis

THICK SKIN

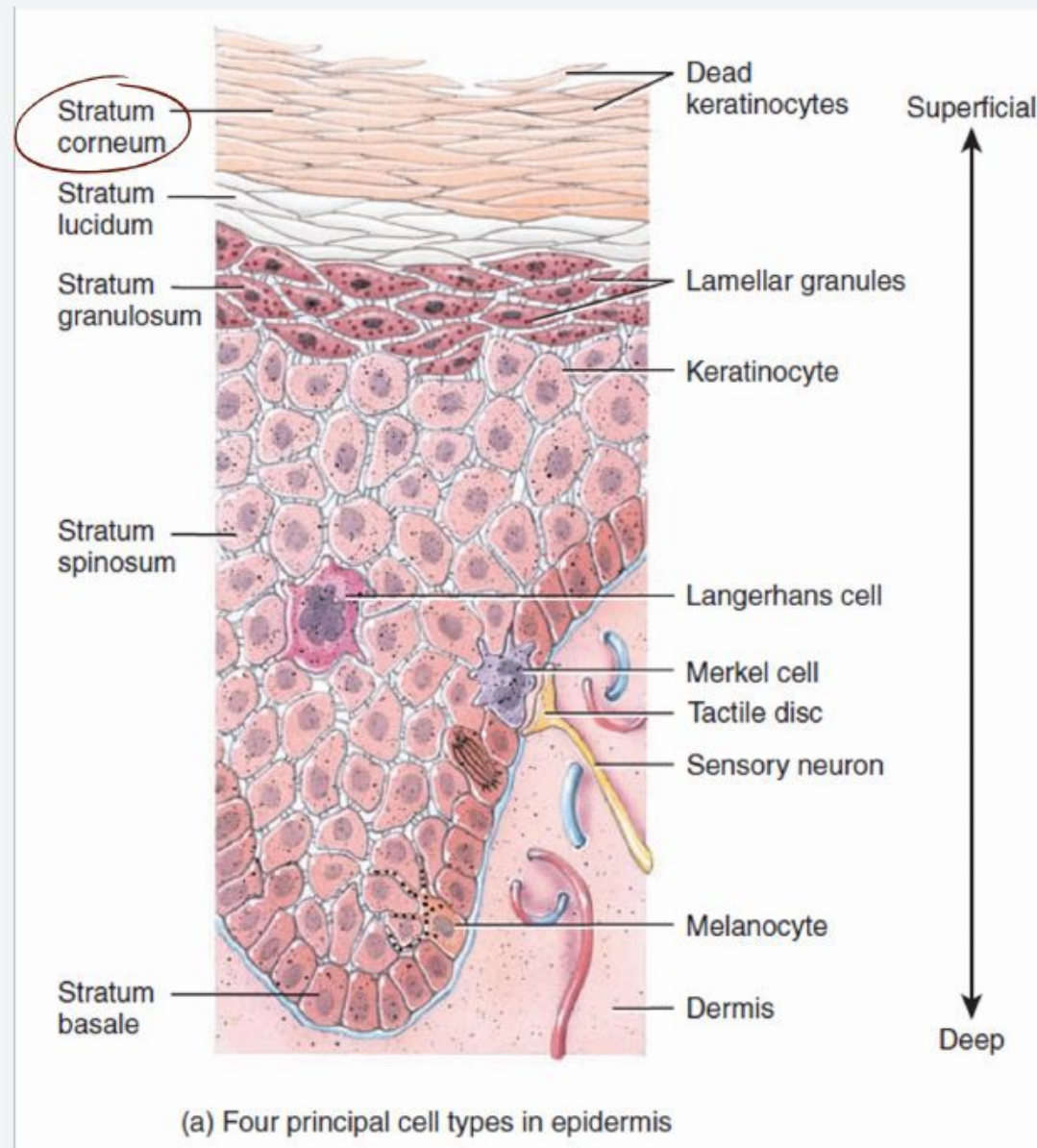
- Stratum Korneum
- Stratum Lucidum
- Stratum Granulosum
- Stratum Spinosum
- Stratum Basalis







EPIDERMIS



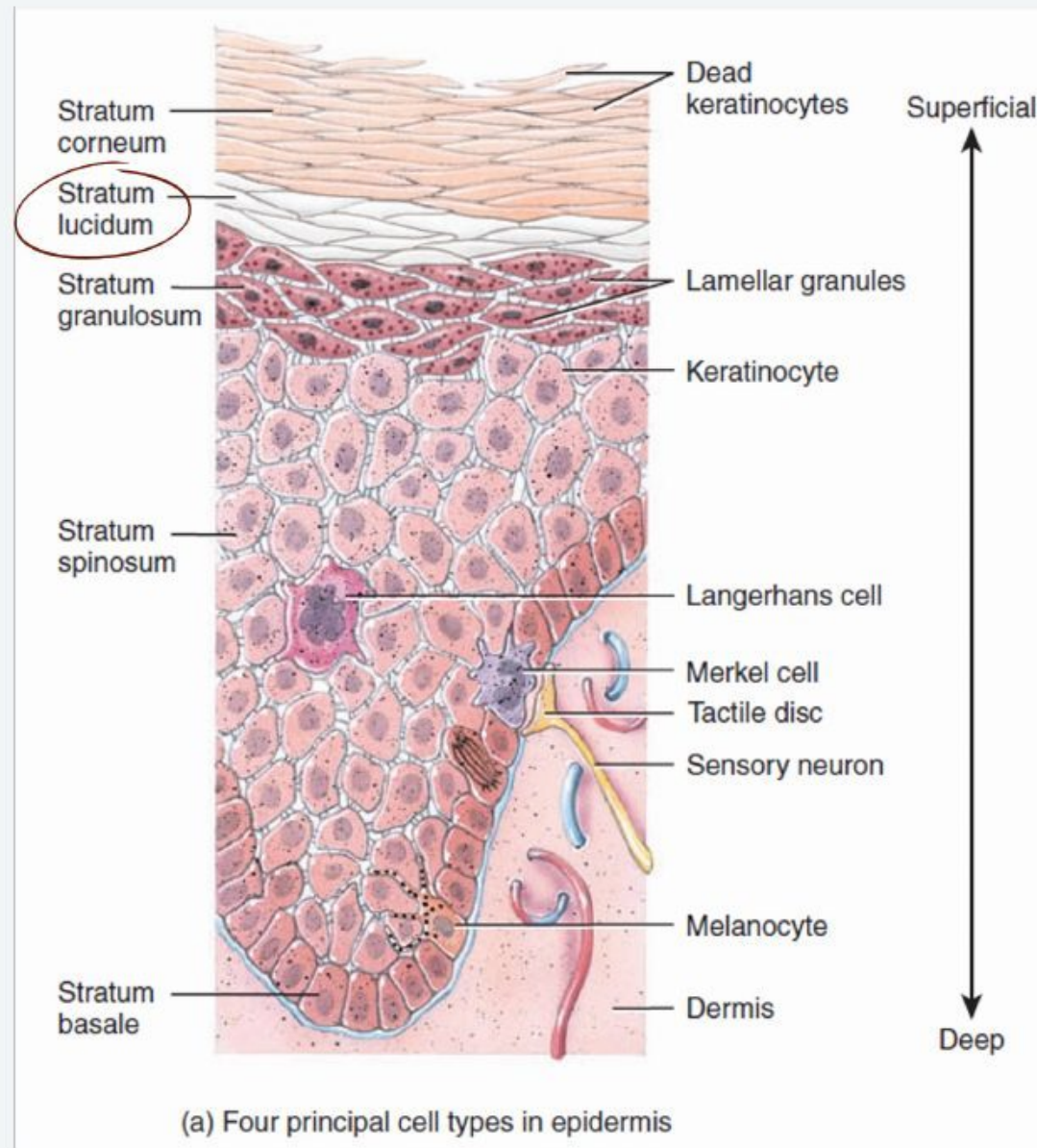
STRATUM KORNEUM

- Consists of 15–20 layers of flattened, dead cells
- The cells are anucleate (lack nuclei)
- The cytoplasm is filled with refractile filamentous keratin
- The superficial portion undergoes desquamation (shedding) as part of keratinization





EPIDERMIS



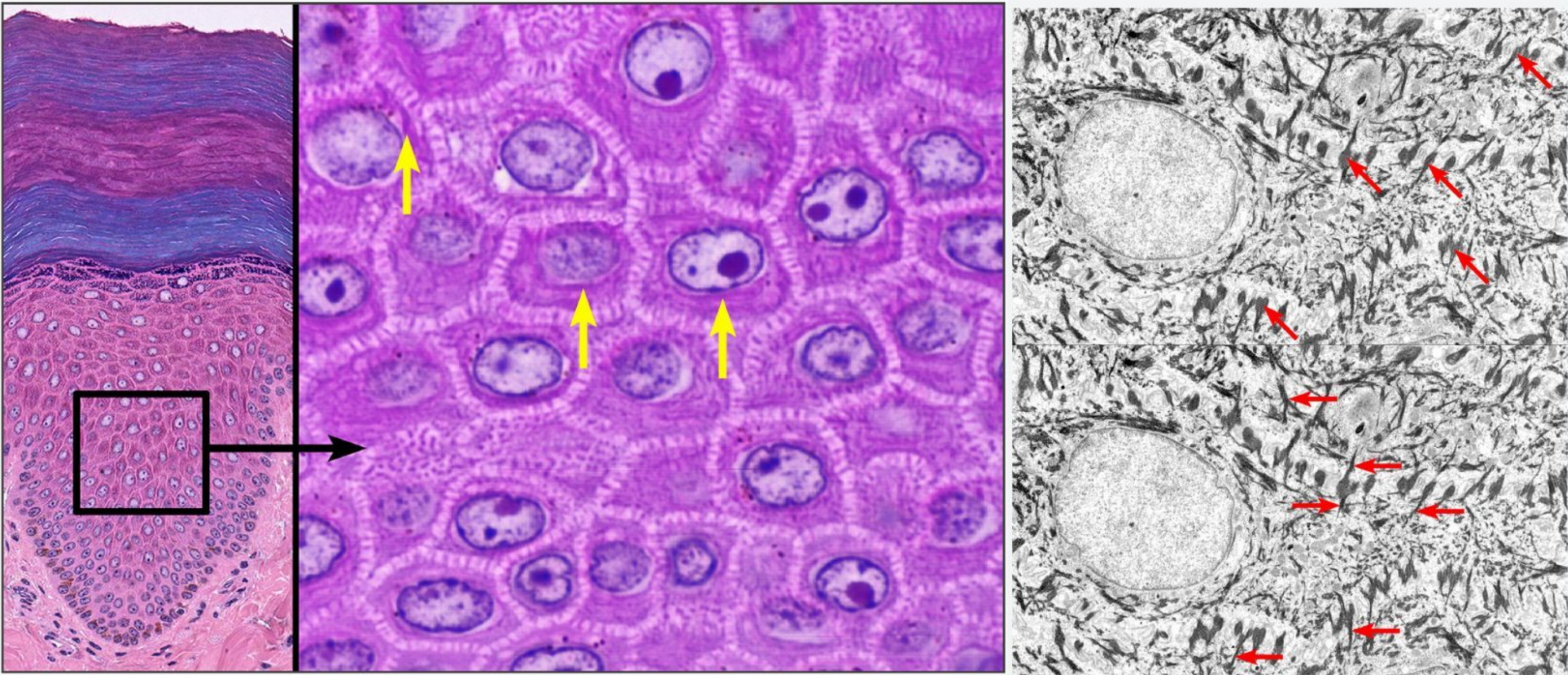
STRATUM LUCIDUM

- Consists of several layers of flattened, pale cells
- Cells are anucleate
- The cytoplasm contains eleidin droplets

!! FOUND ONLY IN THICK SKIN !!



STRATUM SPINOSUM

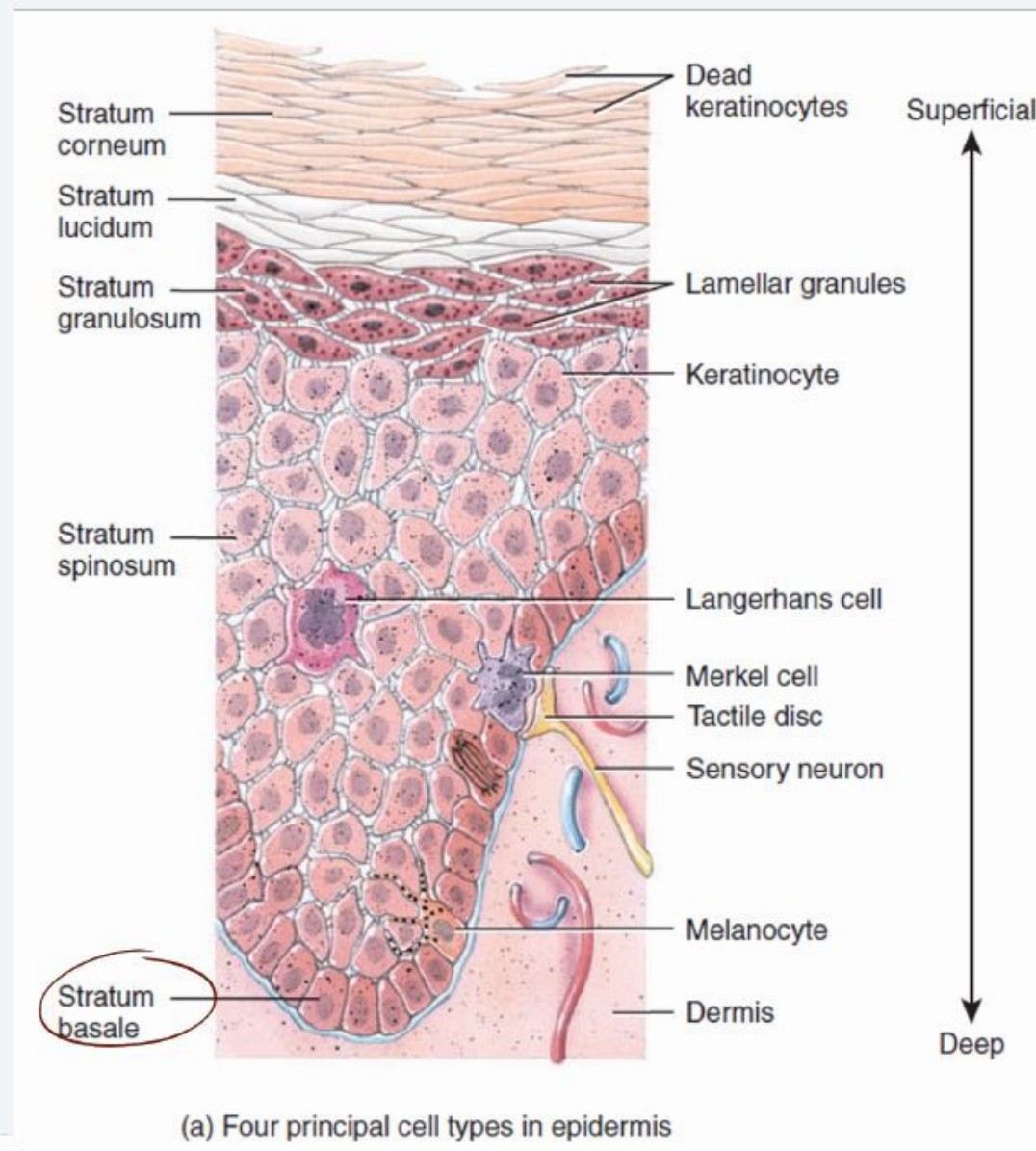


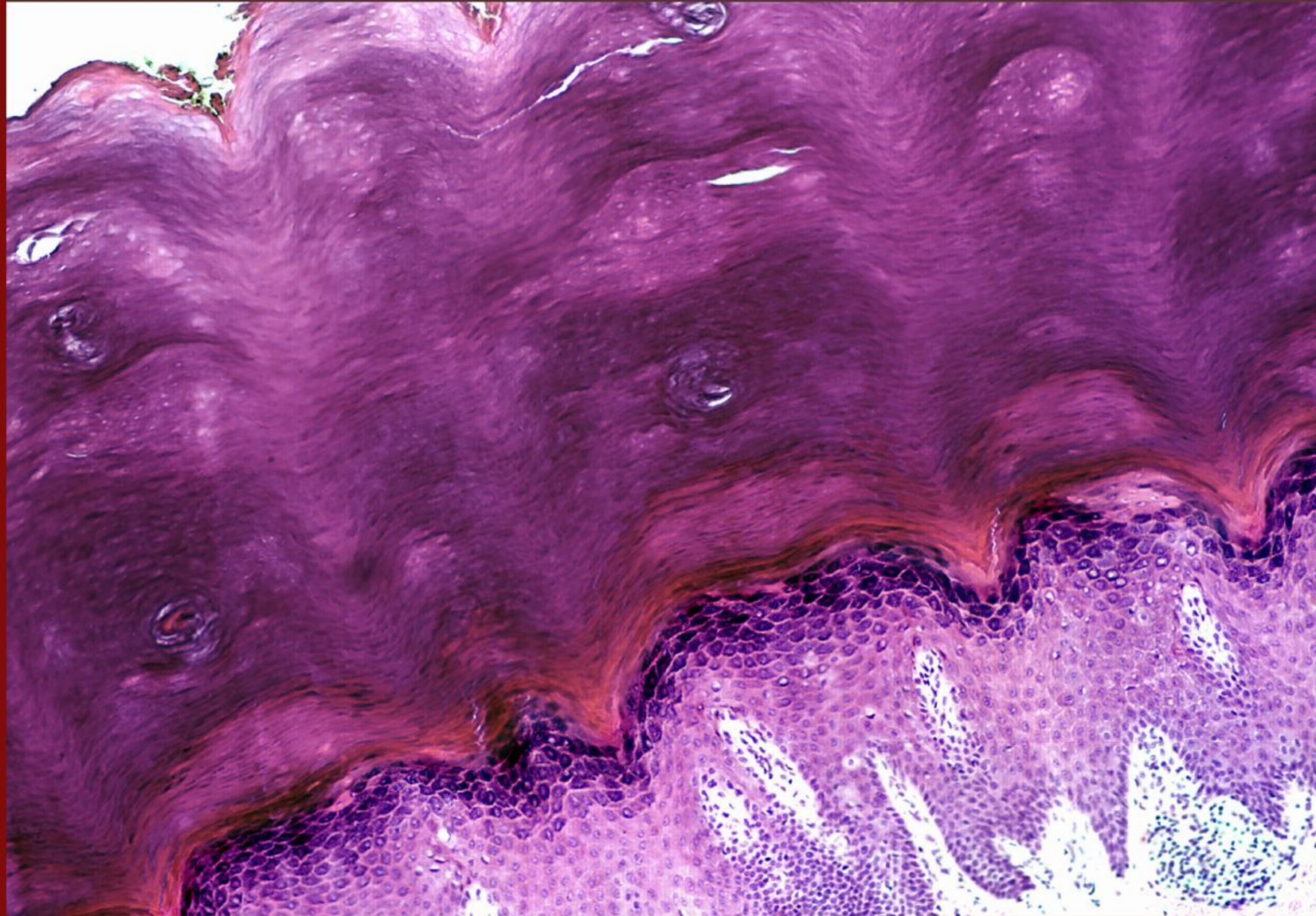


EPIDERMIS

STRATUM BASAL

- A single layer of cuboidal or columnar basophilic cells attached to the basal lamina via hemidesmosomes.
- Possesses desmosomal junctions on the apical and lateral surfaces.
- Has high mitotic activity and contains progenitor cells that give rise to all epidermal cell types.





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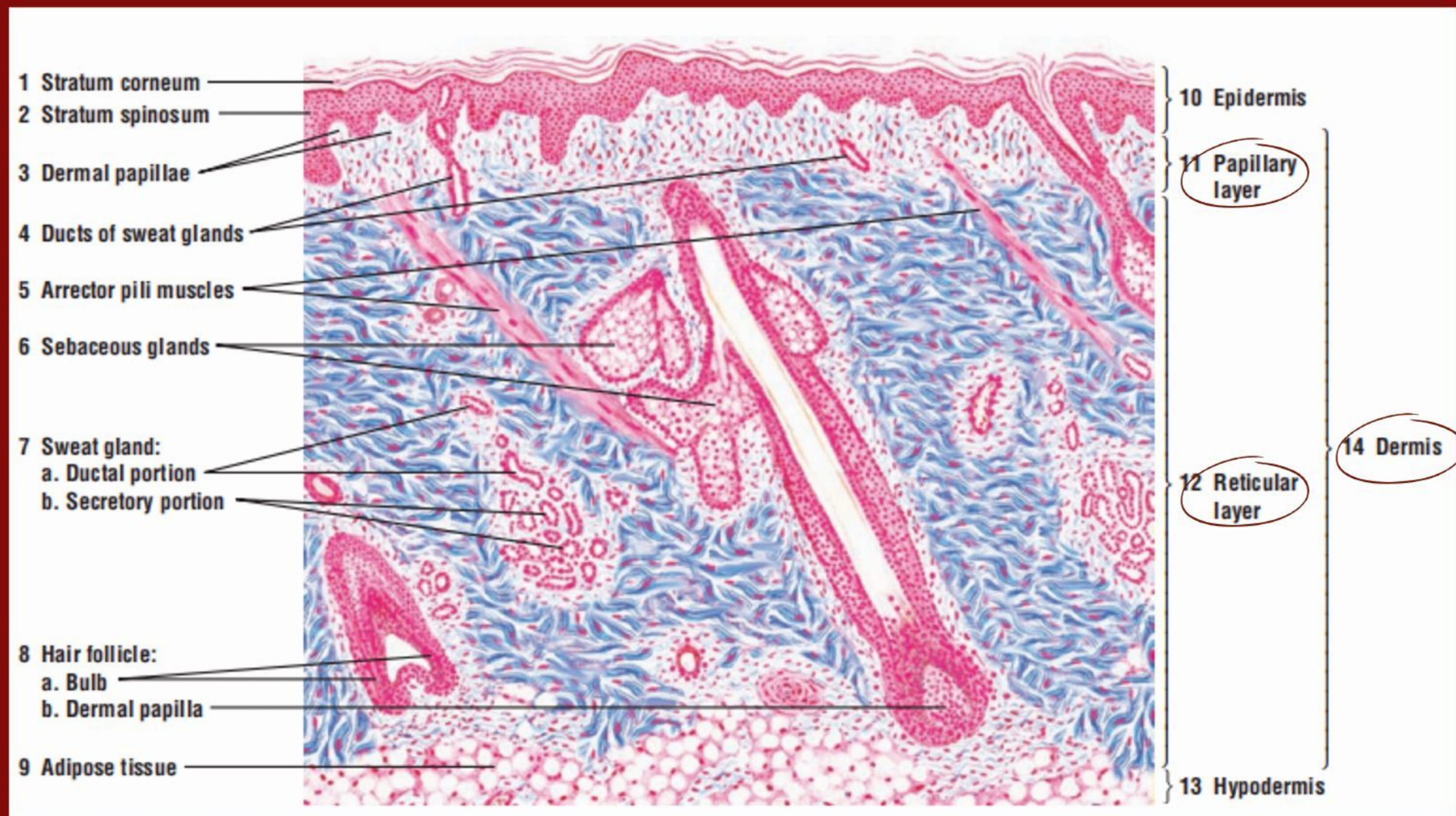
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DERMIS

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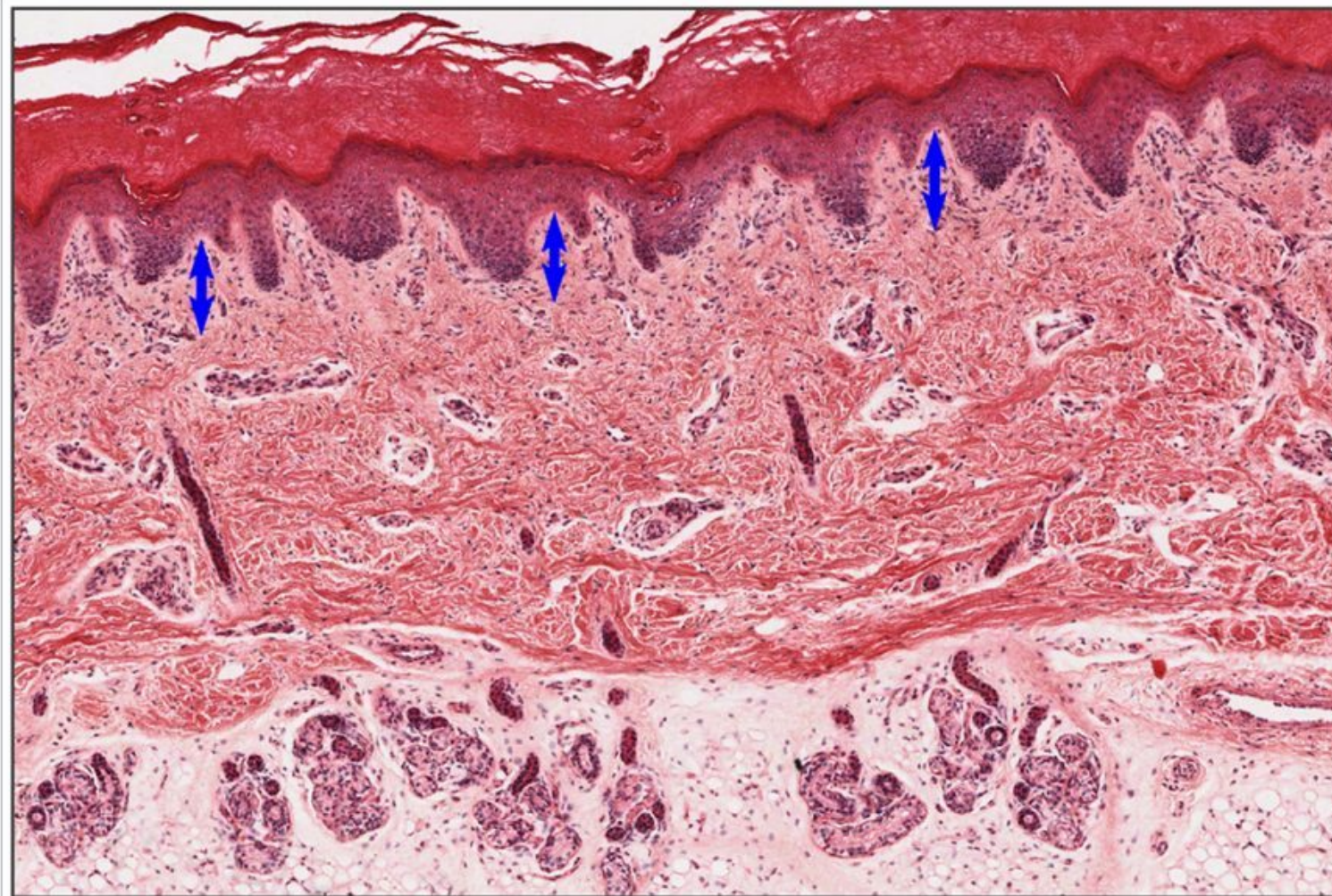


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DERMIS



STRATUM PAPILARE

- A thinner layer
- Composed of loose connective tissue

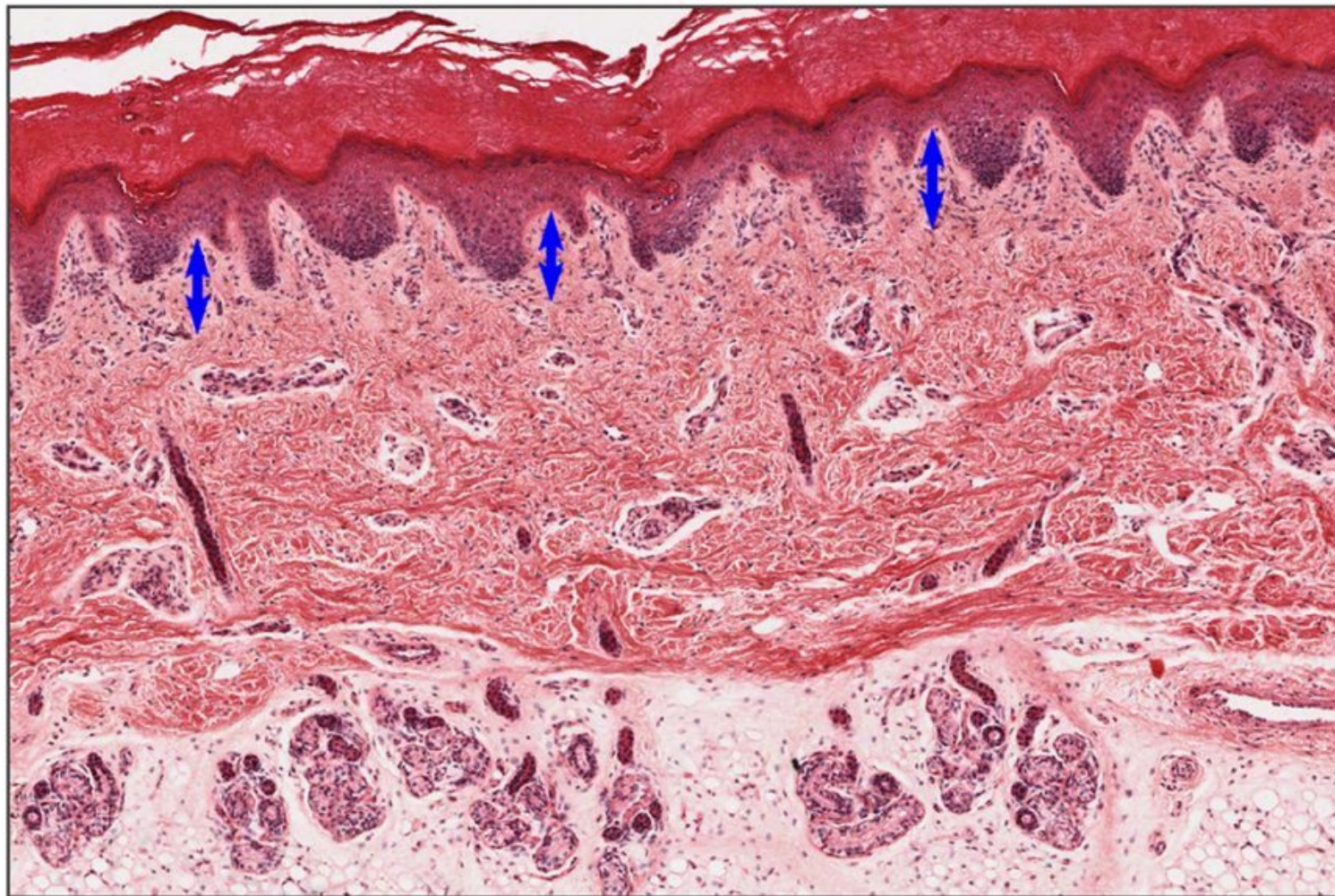
Contains:

- Fibroblasts, macrophages, and mast cells
- Dermal papillae that interdigitate with the epidermal ridges
- Blood vessels that provide nutrients to the epidermis through diffusion
- Anchoring fibrils / type VII collagen
- Meissner's corpuscles





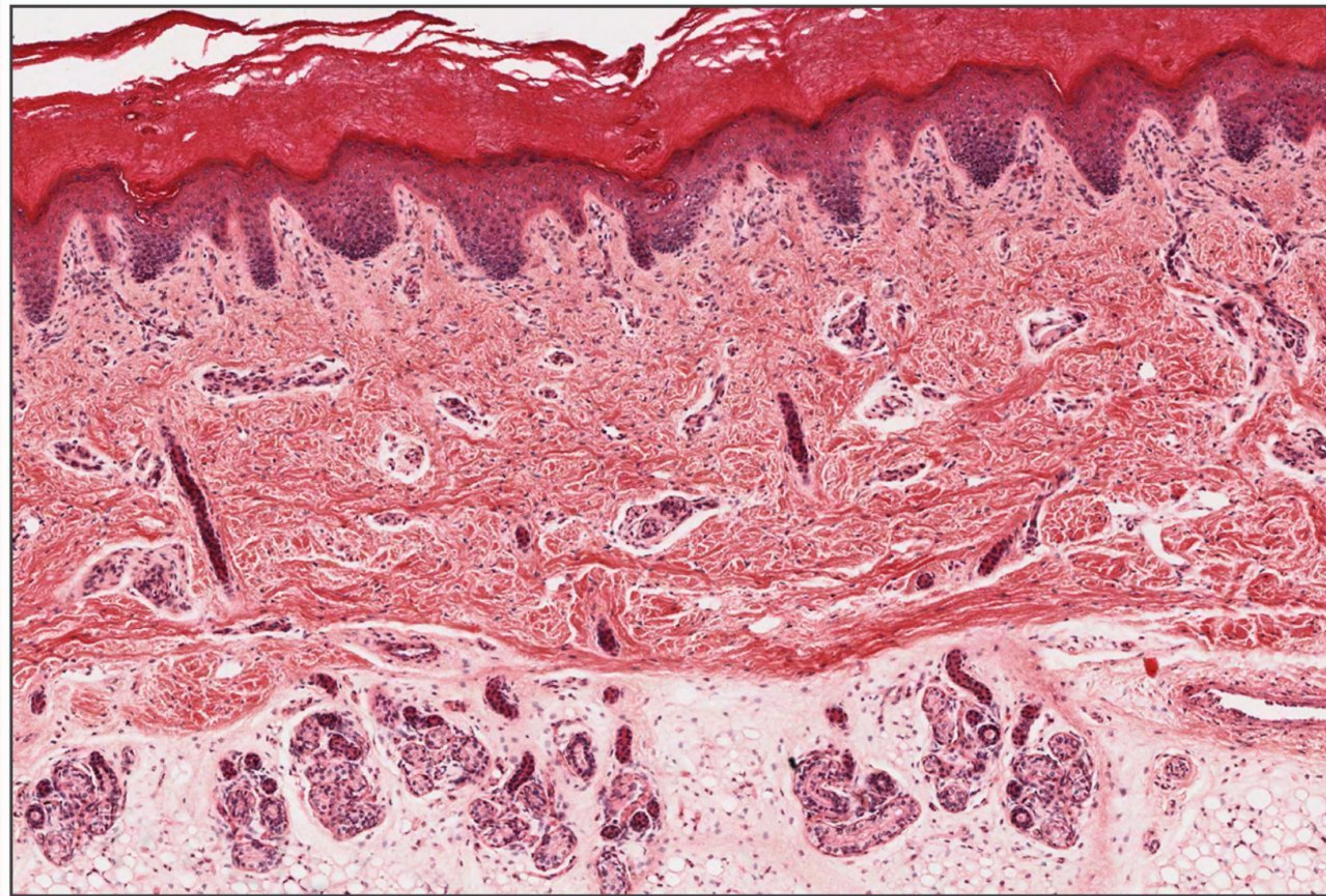
DERMIS



STRATUM RETIKULER

- A thicker layer
- Composed of dense irregular connective tissue, mainly consisting of:
 - Collagen fibers (type I)
 - Elastic fibers





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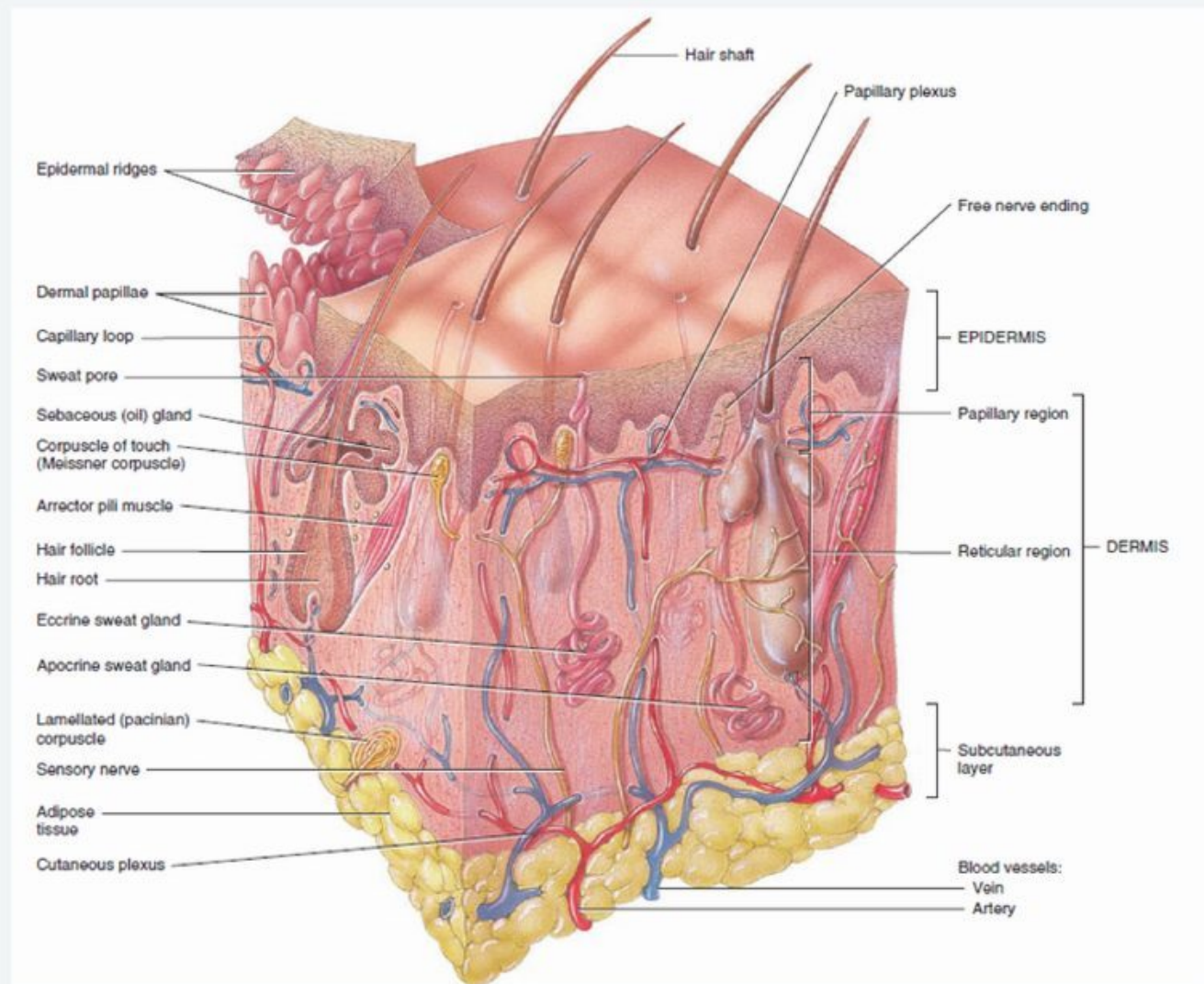
HIPODERMIS

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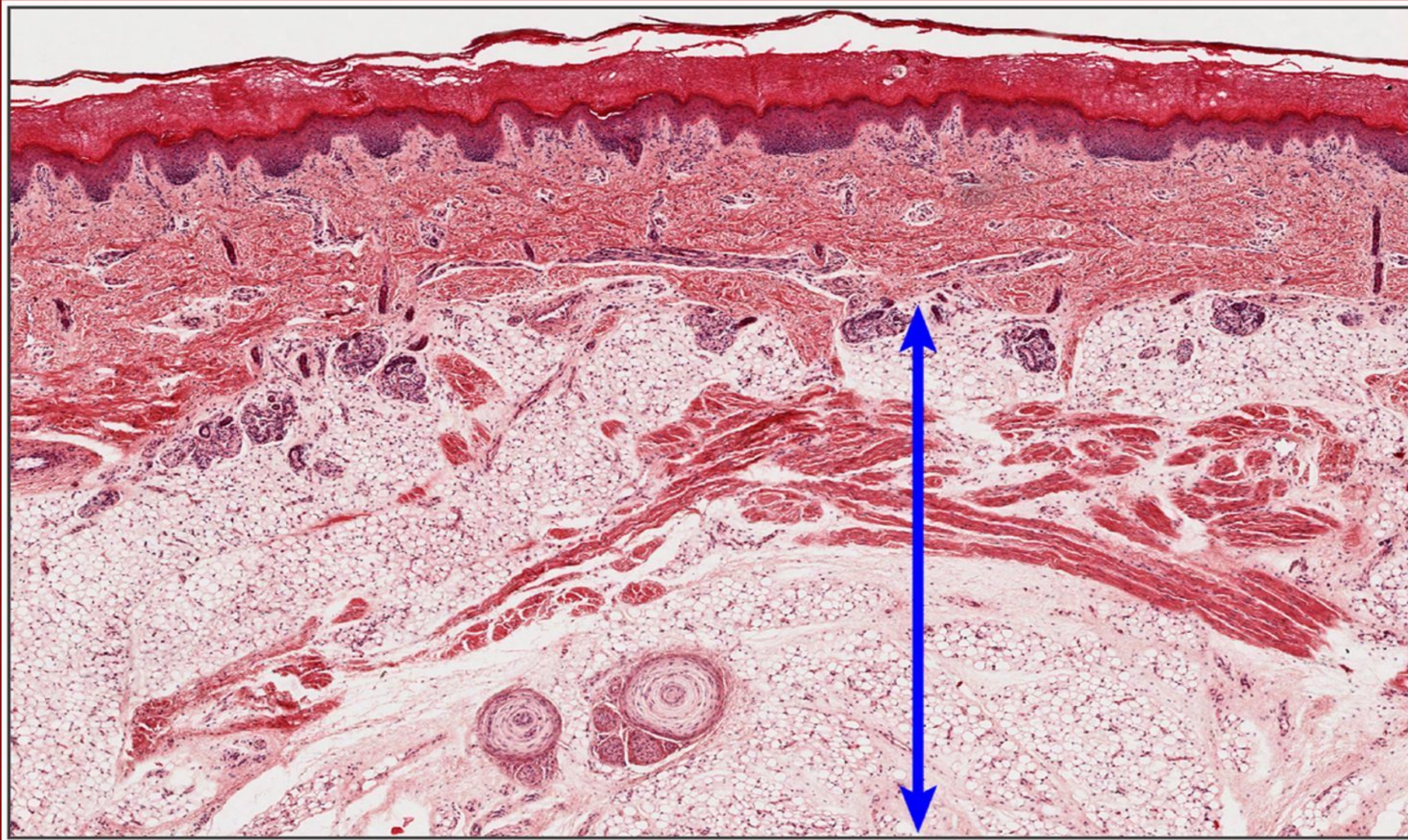


HIPODERMIS



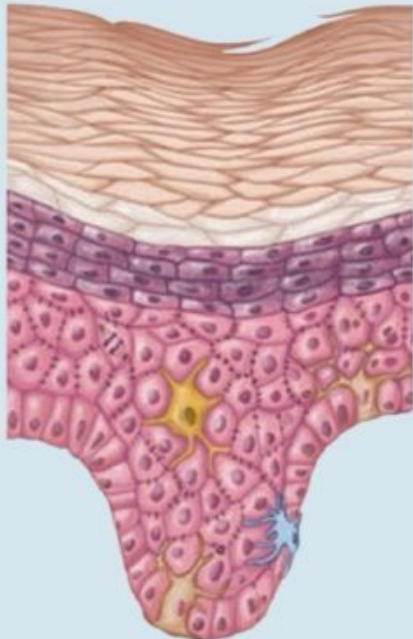
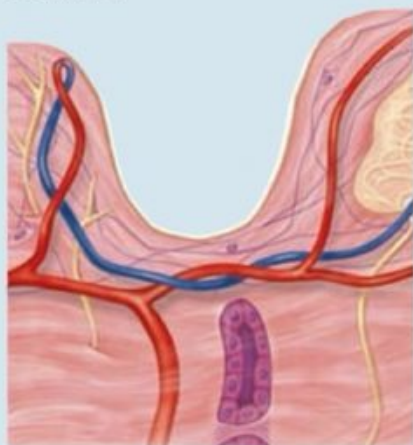
- Composed of loose connective tissue
- Contains abundant adipose cells (adipocytes)
- Has an extensive vascular supply
- Contains Pacinian corpuscles, which are sensory nerve endings that detect pressure stimuli.





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Lapisan	Lapisan Spesifik	Deskripsi
Epidermis		
	Stratum korneum	Lapisan superfisial; 20 sampai 30 lapisan mati, gepeng, nukleat, keratinosit terisi keratin; melindungi terhadap gesekan dan kehilangan cairan
	Stratum lucidum	2 sampai 3 lapis anukleat, sel-sel mati; hanya terlihat pada kulit tebal
	Stratum granulosum	3 sampai 5 lapisan keratinosit dengan granula keratohialin yang berbeda
	Stratum spinosum	Beberapa lapisan keratinosit semua bergabung dengan desmosom; terdapat sel Langerhans
	Stratum basale	Terdalam, lapisan tunggal dari kuboid ke sel kolumnar rendah dalam kontak dengan dasar membran; mitosis terjadi di sini; melanosit dan sel Merkel juga
Dermis		
	Lapisan papiler	Lapisan superfisial dari dermis; terdiri dari jaringan ikat areolar; membentuk dermal papilla; mengandung pleksus vaskular subpapillari
	Lapisan retikular	Lapisan yang lebih dalam dari dermis; jaringan ikat irregular padat di sekitarnya folikel rambut, kelenjar sebacea dan kelenjar keringat, saraf, dan pleksus dalam pembuluh darah memperluas ke lapisan subkutan
Lapisan subkutan	Tidak ada lapisan spesifik	Tidak dianggap sebagai bagian dari integumen; mendalam untuk dermis; terdiri dari jaringan ikat areolar dan adiposa

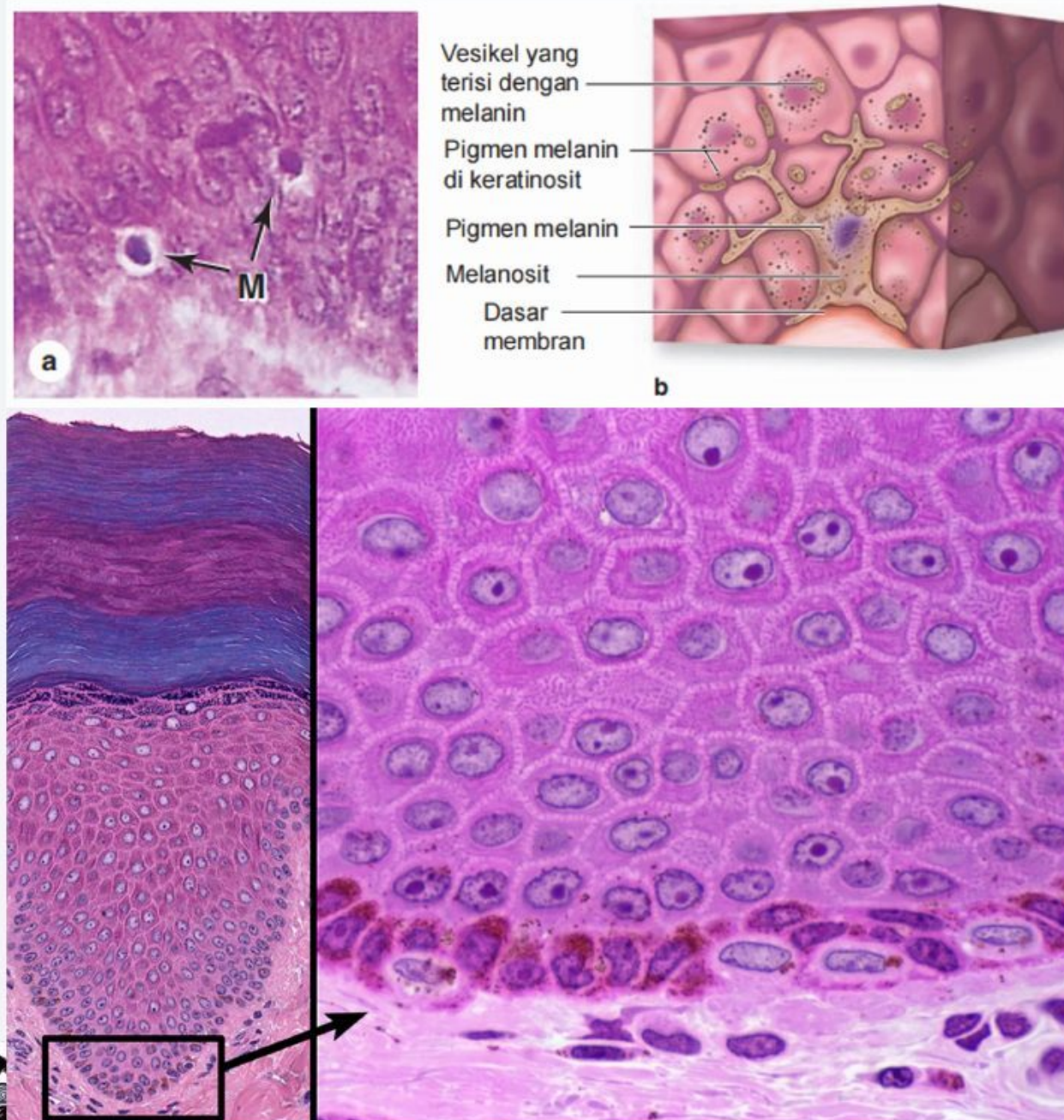
CELLS OF THE EPIDERMIS

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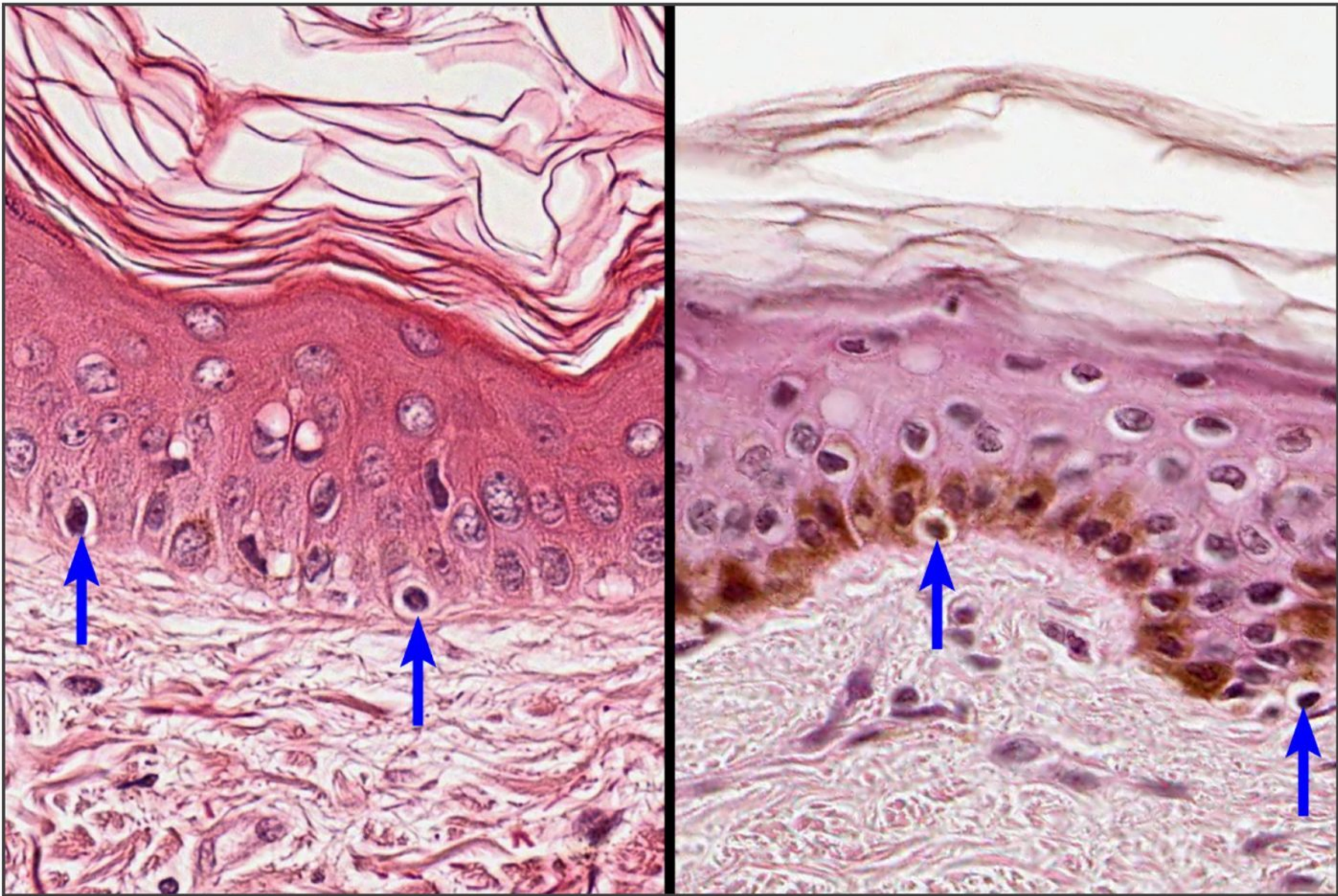


MELANOCYTE



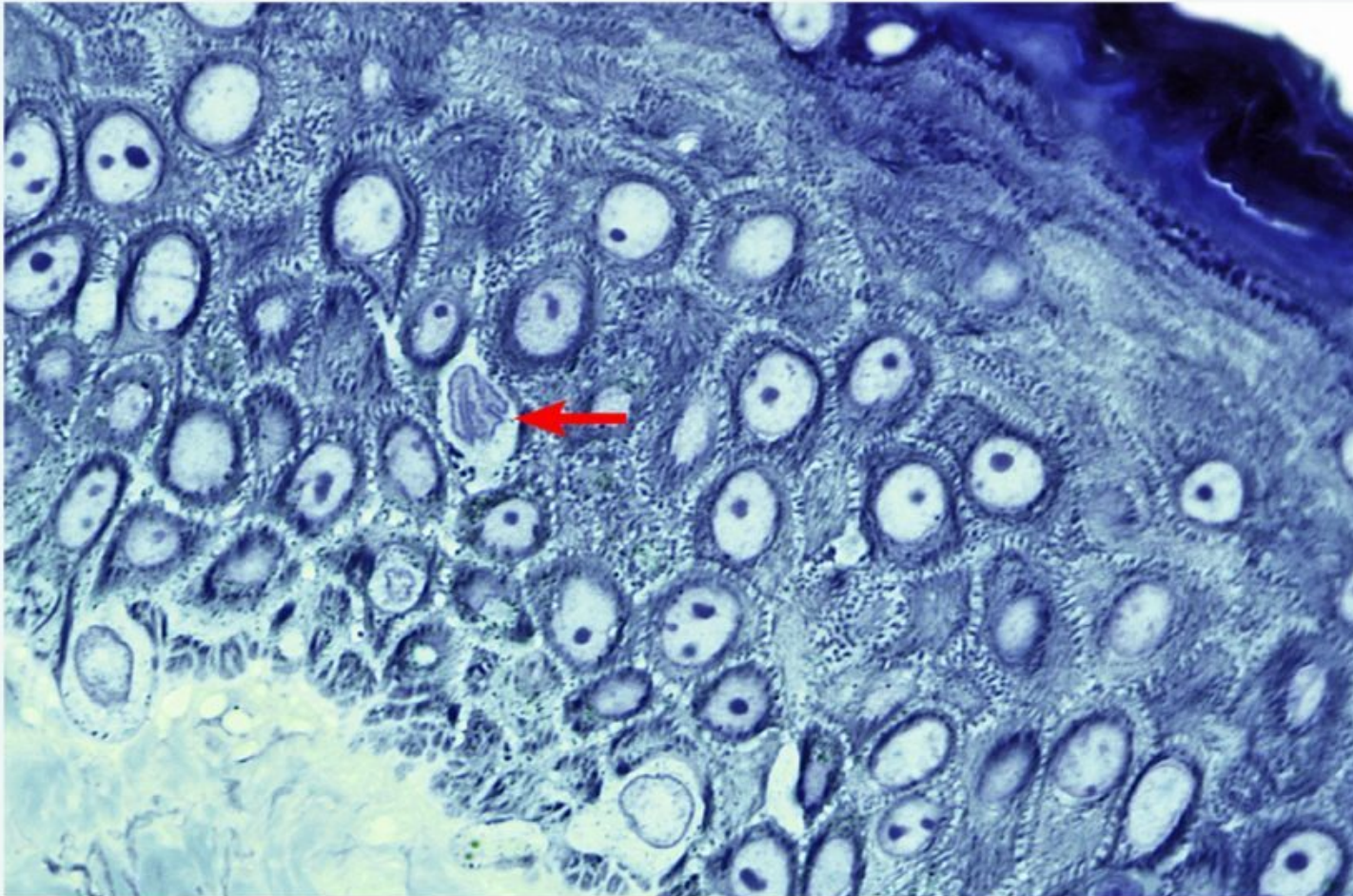
- Round cell body with long cytoplasmic processes
- Located between the cells of the stratum basale
- Functions to produce the pigment eumelanin





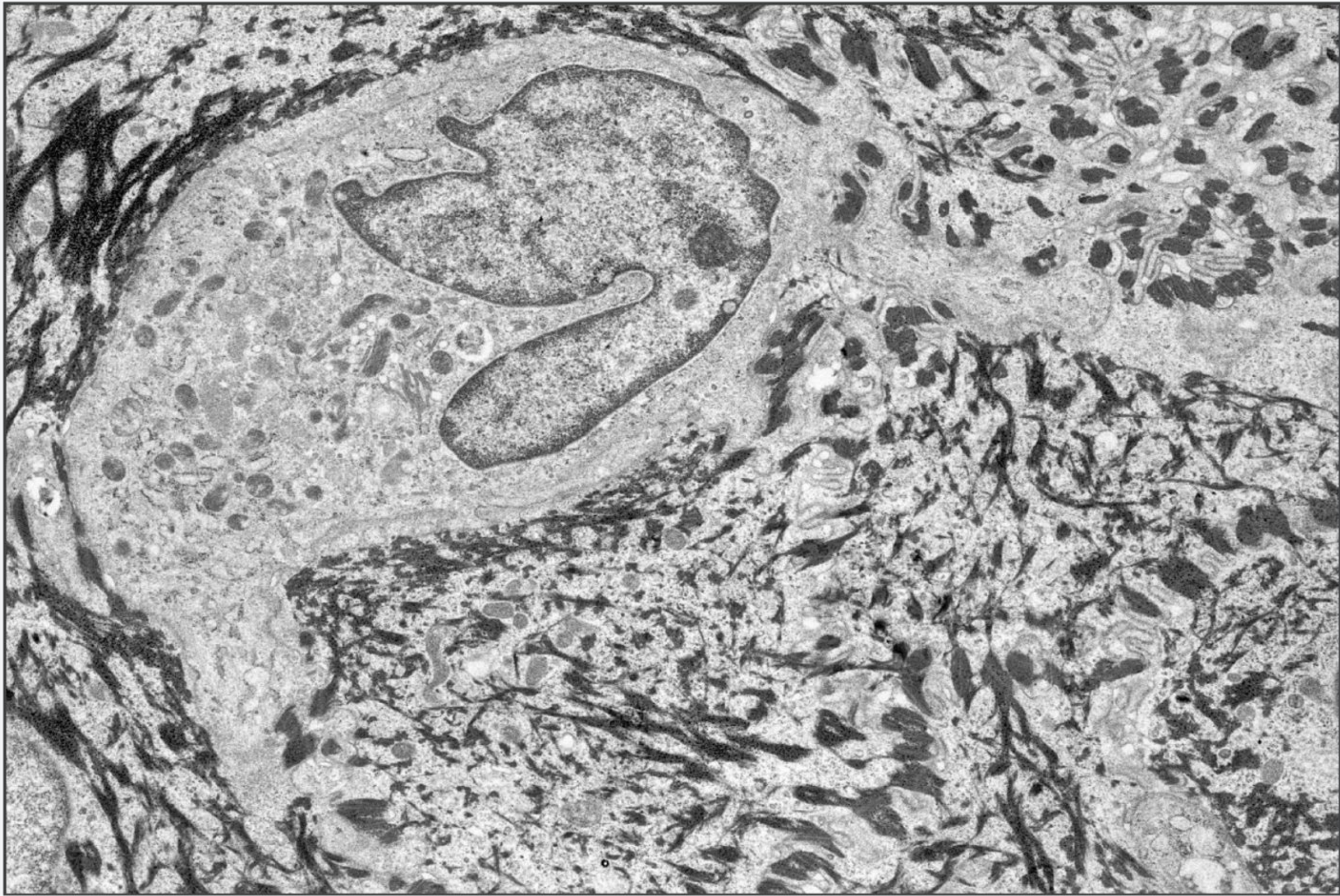


LANGERHANS CELL



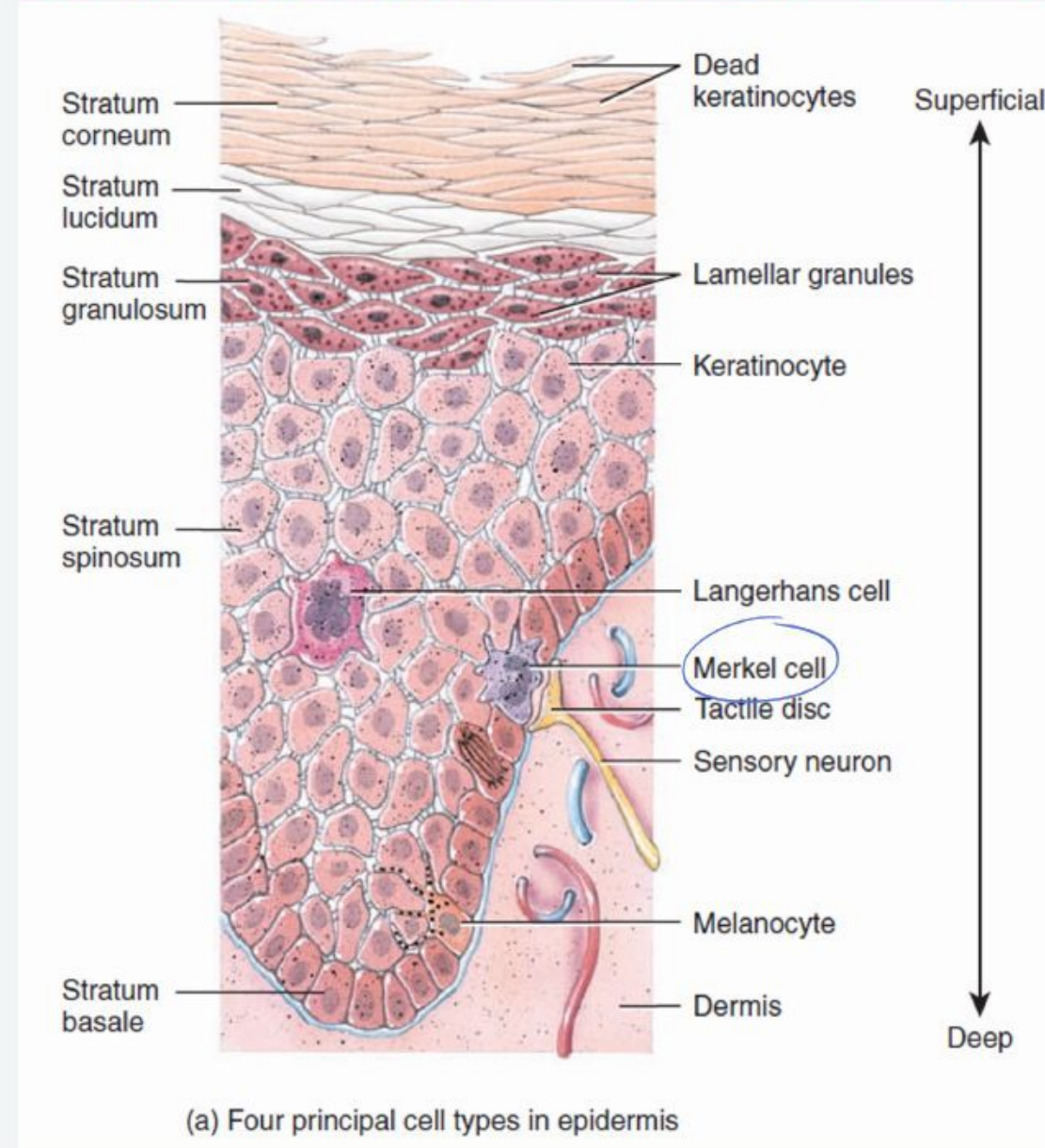
- Morphologically similar to melanocytes, but they do not produce melanin pigment.
- Mainly found in the stratum spinosum.
- Involved in immune activity and function as macrophage-like immune cells.







MERKEL CELL



- Morphologically similar to epidermal epithelial cells, but their cytoplasm contains small, dense granules.
- Generally found in the basal layer of the epidermis, particularly in thick skin.
- Free nerve endings attach to the base of these cells, forming the Merkel corpuscle, which functions as a sensory mechanoreceptor.



LIMPHOCYTE

- Round-shaped cells with a round nucleus containing coarse, densely packed chromatin.
- Most are immature T lymphocytes.
- Their maturation within the epidermis is supported by keratinocytes.
- Together with Langerhans cells and keratinocytes, they form the skin's immune system known as SALT (Skin-Associated Lymphoid Tissue).



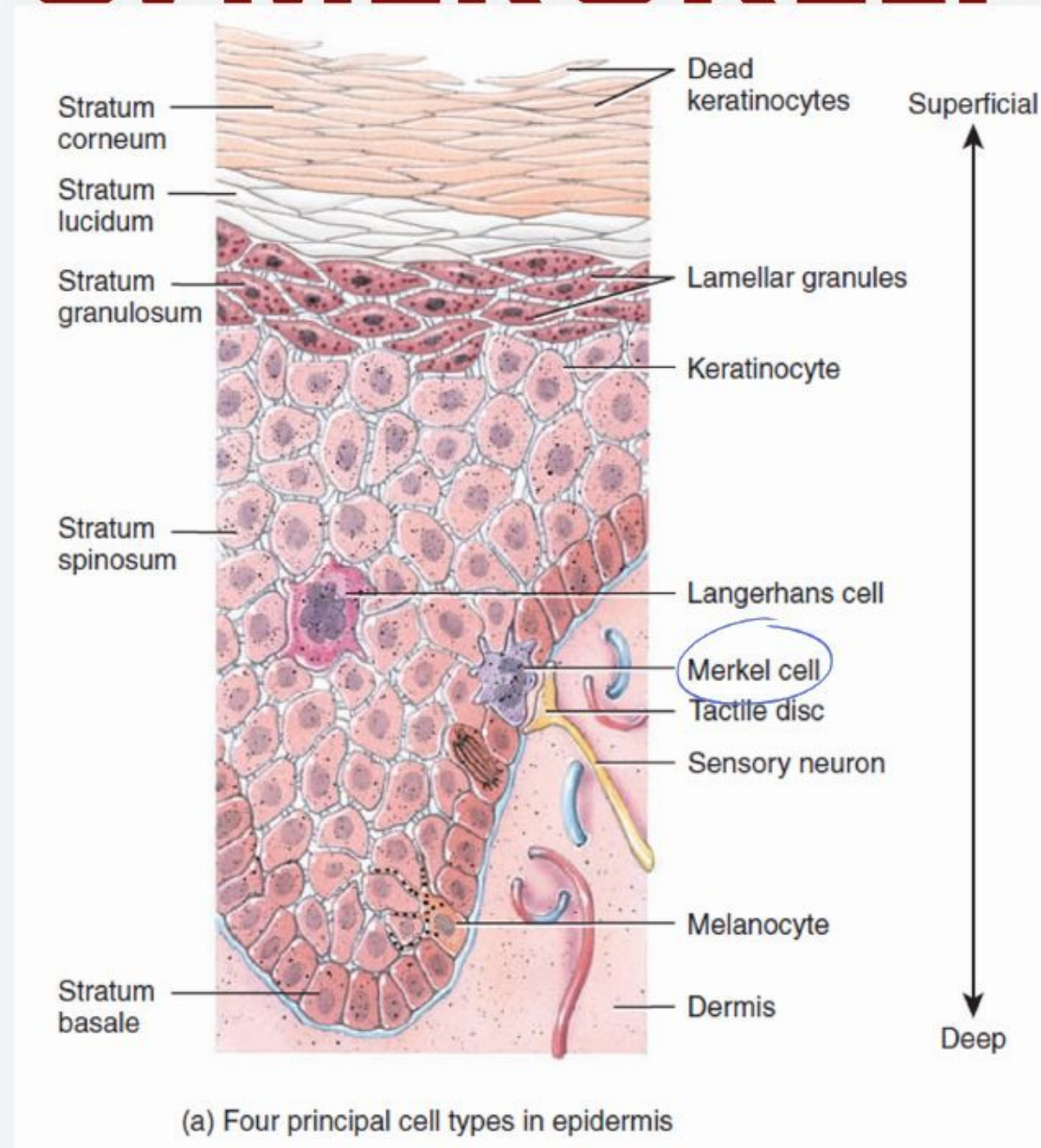
NERVES IN THE SKIN

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C. MERCKELI



- Unencapsulated
- Free nerve endings are attached to Merkel cells
- Located in the basal layer of the epidermis
- Function as receptors for light touch / tactile sensation





C. VATER PACINI

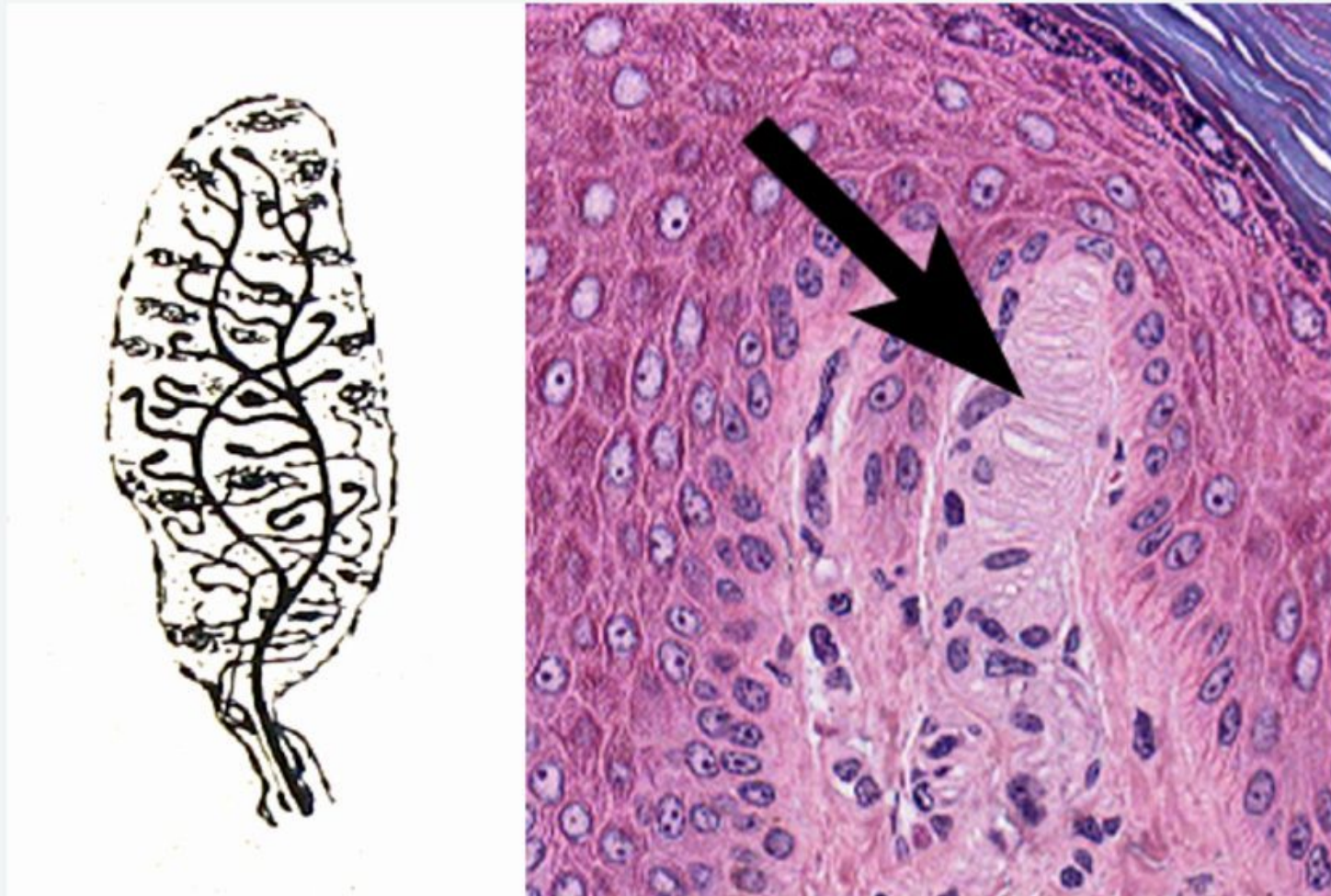


- Oval or onion-shaped
- Nerve endings surrounded by a capsule composed of concentric layers
- The capsule consists of collagen fibers and flattened, endothelial-like cells
- Located in the dermis and subcutaneous tissue
- Functions as a receptor for crude touch, sustained pressure, and vibration





C. MEISSNER



- Cylindrical in shape
- Nerve endings are branched and encapsulated
- Capsule consists of thin connective tissue and transversely arranged flattened cells
- Located in the dermal papillae
- Functions as a receptor for light touch





C. RUFFINI

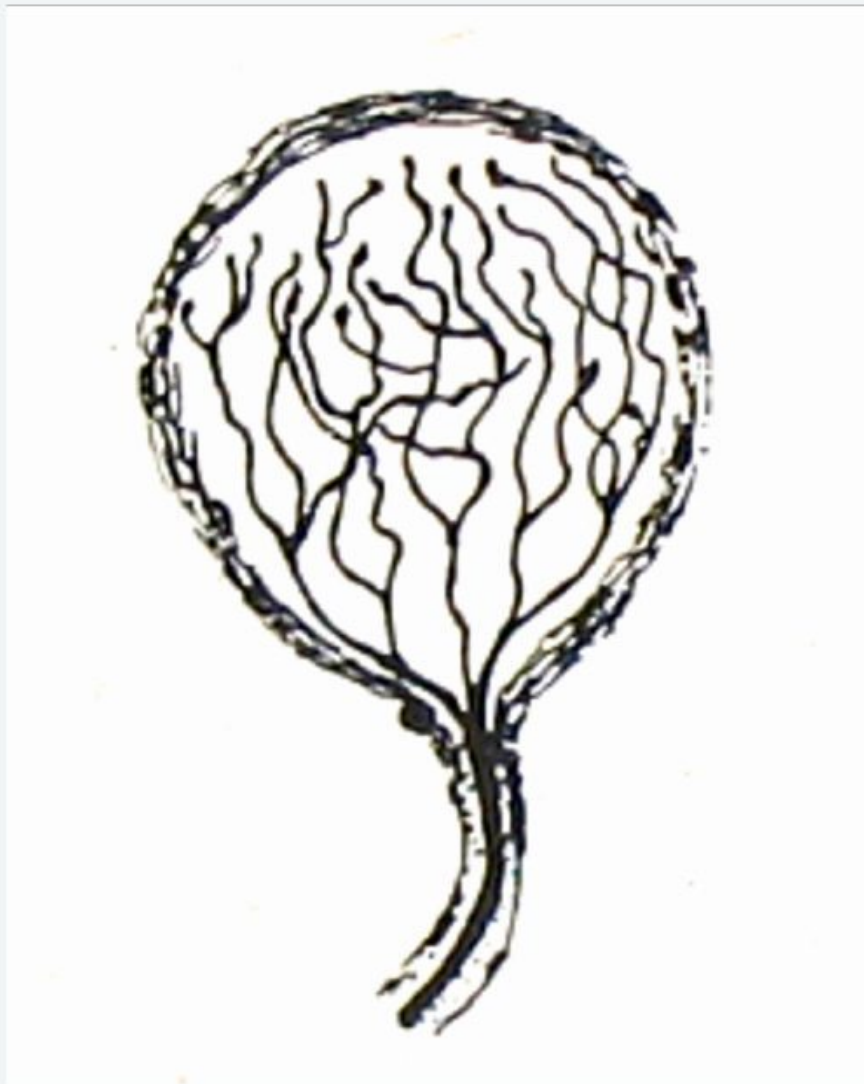


- Cylindrical and smaller than Meissner's corpuscles
- Nerve endings branch in a spray-like pattern and are encapsulated
- The capsule consists of thin connective tissue
- Located in the dermis
- Functions as a receptor for pressure





C. KRAUSE

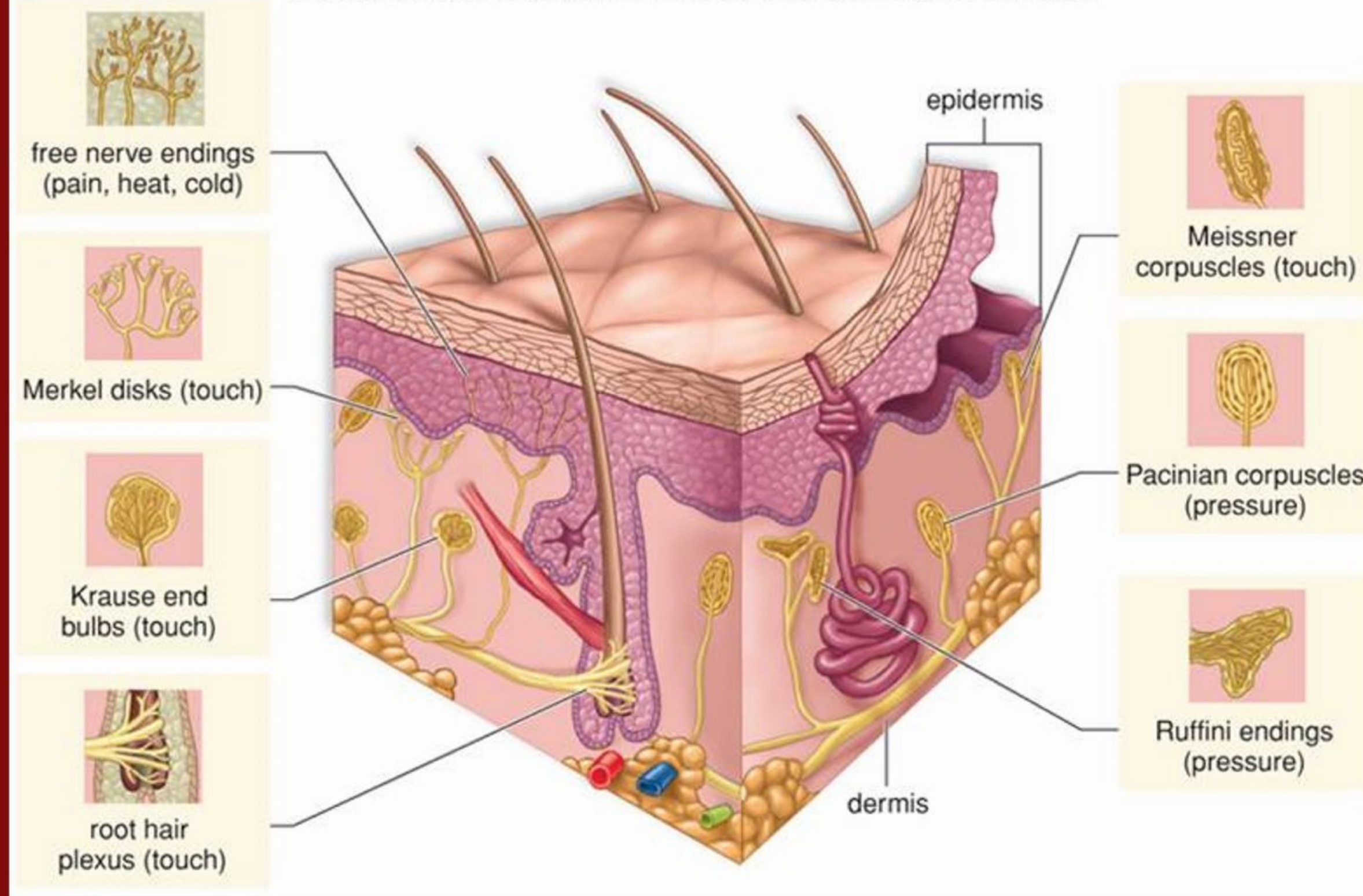


- Round / spherical
- Nerve endings are branched and encapsulated
- Capsule consists of thick connective tissue
- Located in the dermis
- Functions as a receptor for pressure





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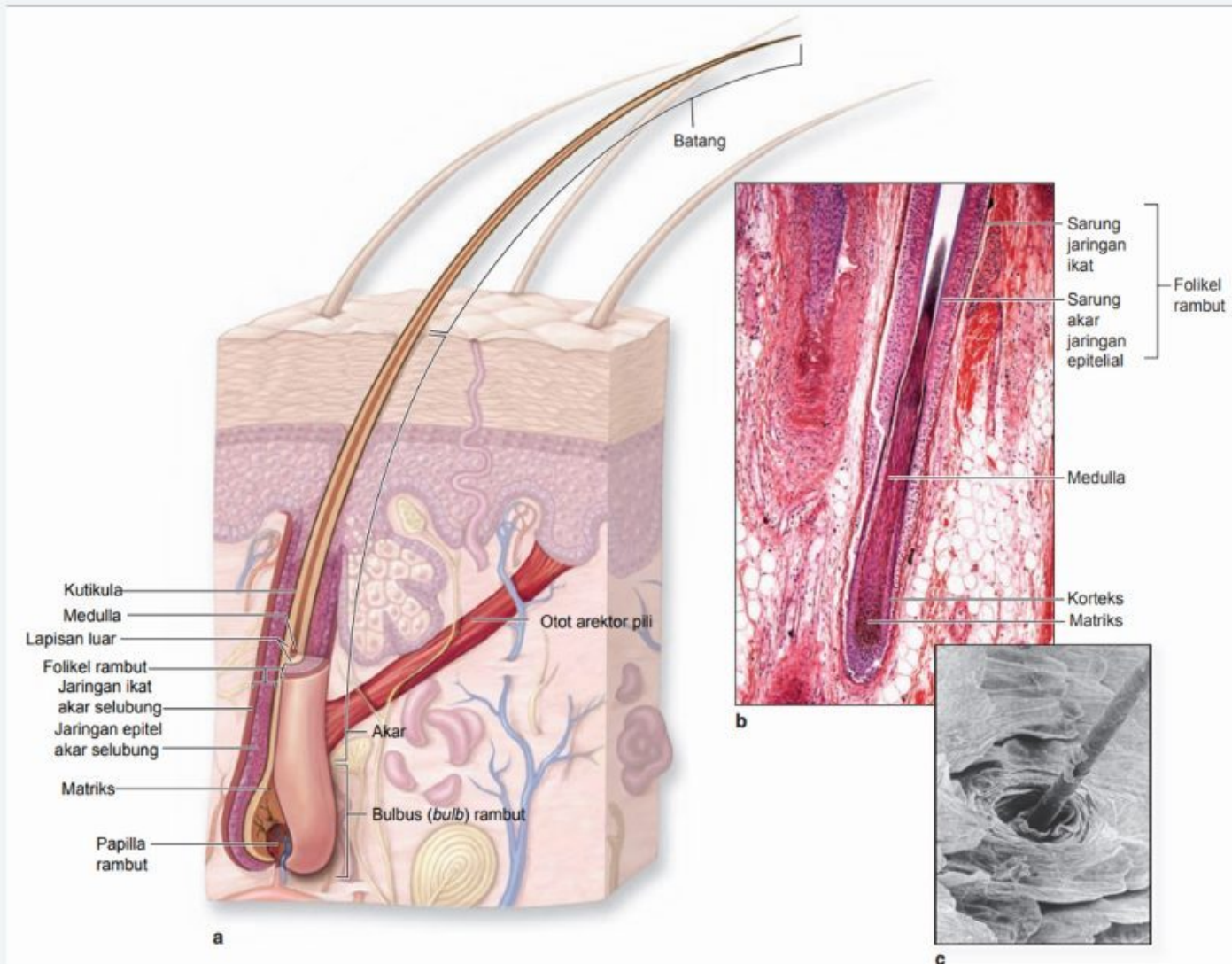
SKIN APPENDAGES

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HAIR

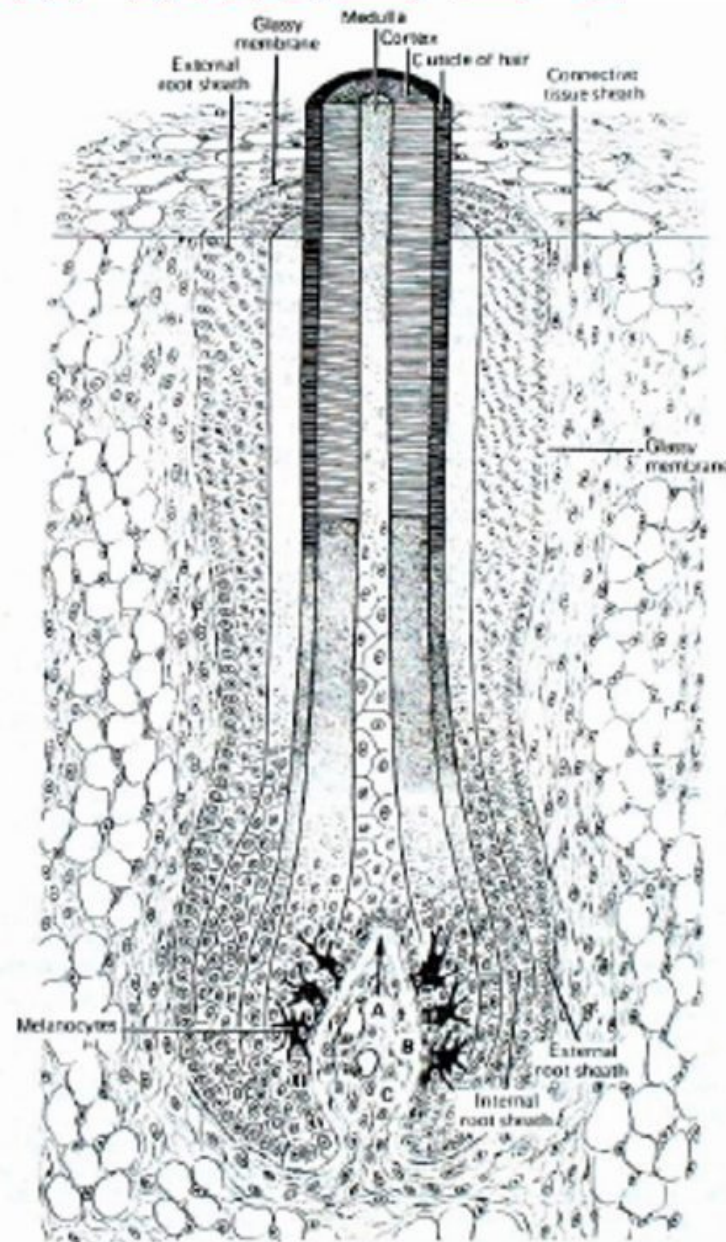


Hair is divided into:

- Hair shaft
- Hair root → enclosed by the hair follicle
- Hair bulb → an expanded portion of the hair follicle, containing:
 - Dermal papilla composed of connective tissue from the dermis
 - Numerous blood vessels
 - Epithelial cells as the site of regeneration
 - Melanocytes located between the epithelial cells



RAMBUT



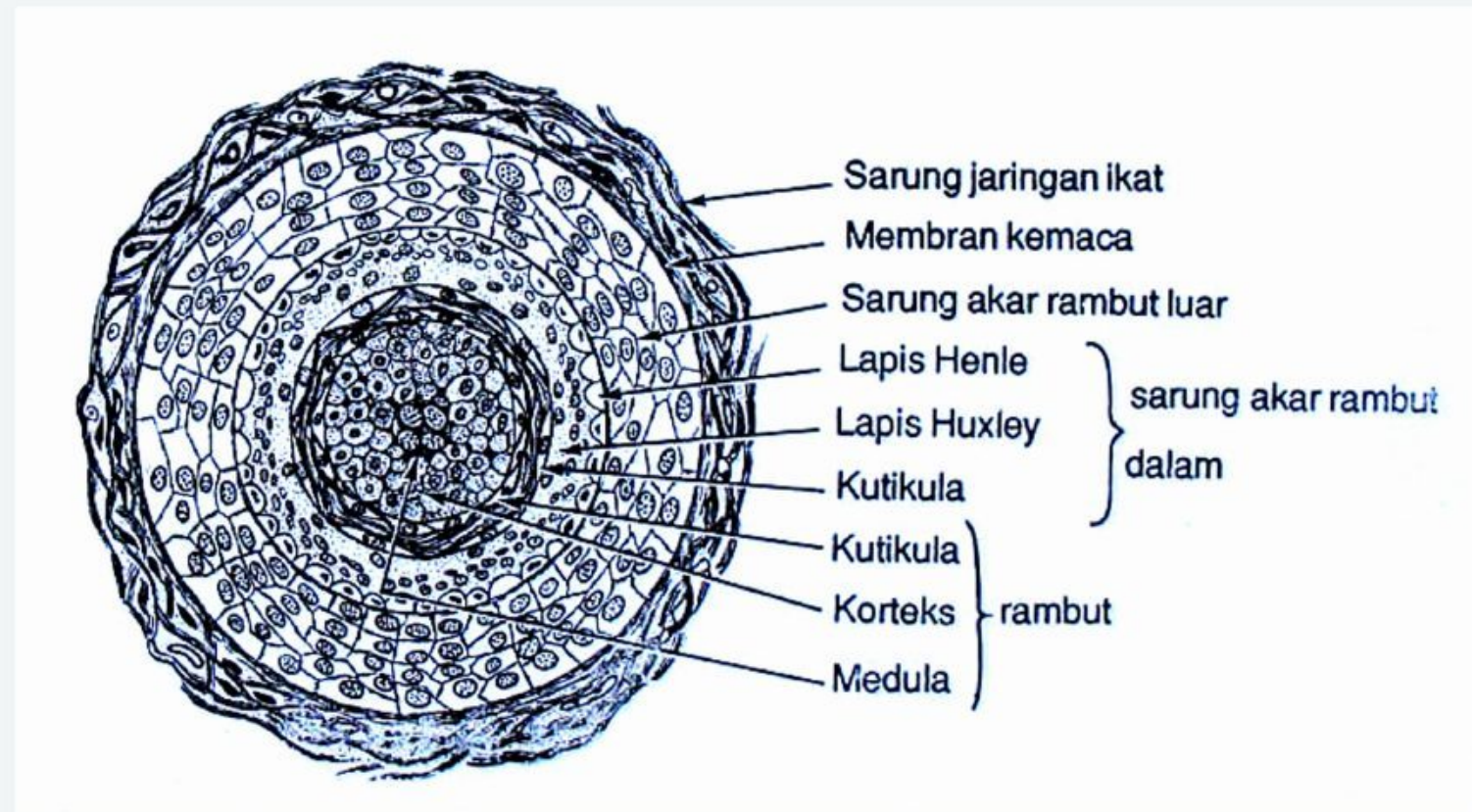
consists of three concentric layers from the inside to outside:

1. Medulla
2. Cortex
3. Cuticle





RAMBUT



In general, the sheath consists of two main parts, separated by the glassy membrane:

- Dermal Root Sheath
 - The outer layer, composed of connective tissue similar to the dermis.
 - Serves as the attachment site for one end of the arrector pili muscle.
- Epidermal Root Sheath
 - The inner layer, composed of epithelial tissue similar to the epidermis.

